

**AUSM LAB** York University · Lassonde School of Engineering

POSTDOCTORAL OPPORTUNITIES

# Postdoctoral fellows

## AI world models for cities, Earth and space

Toronto, Canada

### Urban World Models

City-scale robot–mobility interaction; graph models that understand and predict the city

### Scan-to-Model

Infrastructure-knowledge-compatible 3D models from scans

### Embodied Autonomy

Robots, drones and self-driving vehicles that sense, map and act in real time

### Physics-Informed Earth AI

Physics-informed AI for Earth emulation

### Space Situational Awareness

Space-in-the-loop sensing, detection and tracking

Start
**Flexible (2026–2027)**

Applications
**Reviewed on a rolling basis**

Email your **CV, research statement and 2–3 papers** to **gsohn@yorku.ca**
Subject: **[Postdoc] Your Name**

ausmlab.com/join

POSTDOCTORAL OPPORTUNITIES

# Postdoctoral fellows

**Applications reviewed on a rolling basis · Start: Flexible (2026–2027)**

Build AI world models that understand and predict cities, infrastructure, the Earth and near-Earth space. Lead a research thrust, mentor graduate students and work directly with industry and government partners.

## ABOUT THE LAB

The AUSM Lab at York University develops embodied spatial intelligence: AI systems that perceive, reason about and act within physical space. Our work spans urban mobility and robotics, infrastructure digital twins, autonomous mapping, and AI for Earth and space.

We are looking for postdoctoral fellows to lead one of five research thrusts. You will work at the centre of a team of Ph.D. and M.Sc. students, and with industry and government partners such as Teledyne Optech, Aethon Aerial Solutions and the Ontario Ministry of Transportation.

## RESEARCH THRUSTS

- **Urban World Models**
  City-scale robot–mobility interaction; graph models that understand and predict the city
- **Scan-to-Model**
  Infrastructure-knowledge-compatible 3D models from scans
- **Embodied Autonomy**
  Robots, drones and self-driving vehicles that sense, map and act in real time
- **Physics-Informed Earth AI**
  Physics-informed AI for Earth emulation
- **Space Situational Awareness**
  Space-in-the-loop sensing, detection and tracking

## WE OFFER

- Funding through NSERC Discovery, NSERC CREATE SMART and industry partners
- Support to apply for the Canada Postdoctoral Research Award (CPRA) and Mitacs Accelerate fellowships
- Experience leading industry projects and mentoring graduate students
- A GPU computing cluster and real-world data from industry partners

## YOUR ROLE

- Lead a research thrust, from problem definition to publication
- Publish in top venues (CVPR, ICCV, ECCV, ISPRS, IEEE TGRS)
- Co-supervise Ph.D. and M.Sc. students day to day
- Run industry projects and meet with partners
- Help write grant and fellowship proposals

## WHO WE LOOK FOR

- A Ph.D. (completed or near completion) in computer vision, machine learning, robotics, photogrammetry and remote sensing, geomatics, or a related field.
- A strong publication record in leading venues (e.g., CVPR, ICCV, ECCV, ISPRS, IEEE TGRS).
- Hands-on experience with deep learning (Python, PyTorch) and 3D or spatial data.

## HOW TO APPLY

Email **Prof. Gunho Sohn, gsohn@yorku.ca** with:

- Your CV with a full publication list
- A research statement (up to two pages)
- Two or three representative publications
- Names and contact details of three referees

Subject line: **[Postdoc] Your Name**

## FUNDING

Positions are funded through the lab's research grants and industry partners. We also encourage strong candidates to apply with us for external fellowships, such as the Canada Postdoctoral Research Award (CPRA) and Mitacs Accelerate, and we support the application.